Viewpoints in Complex Event Processing

Industrial Experience Report

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ABSTRACT

In the automation industry an increasing need for integration exists, spanning from field level to enterprise resource planning. An important requirement is the adaptability of the involved production processes, because industry must be able to react quickly to new situations and challenges. Complex Event Processing (CEP) is a promising technology to detect and react to such situations. However, current approaches for CEP often mix the derivation of high-level business events with their interpretation and follow a bottom-up development that is not focused on Key Performance Indicators (KPIs). Consequently, maintainability and the required adaptability are hard to achieve.

In this paper, we investigate the separation of two viewpoints in a CEP application: i) modeling critical business situations with the reactions to be taken and ii) aggregation of the necessary base data to distill key performance indicators. Additionally, we present a methodology for the engineering of these viewpoints. This gives guidance for the development of the underlying eventing infrastructure and provides a goal/KPI-oriented approach.

Categories and Subject Descriptors

D.2.2 [**Software Engineering**]: Design – *methodologies, representation;* D.2.10 [**Software Engineering**]: Design Tools and Techniques – *top-down programming;* J.7 [**Computers in other Systems**]: *Industrial control, process control*

General Terms

Management, Measurement, Performance, Design, Economics.

Keywords

Complex Event Processing, engineering, methodology, business situation, event processing network

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1. INTRODUCTION

The automation industry is facing increasing competition as transport costs sink, organizational boundaries are reshaped often, and the integration of business processes proliferates. Business agility is a driving force that requires prompt reactions to critical situations and the ability to adapt whole processes in order to realize opportunities and to prevent losses. Examples for such situations are degrading precision in a manufacturing process or a shortage of incoming goods. In both cases, quick reactions are necessary to prevent severe problems in later production steps.

Complex Event Processing (CEP) is seen as an important technology to address these situations [10]. Discussions about the benefits of CEP usually begin with the need for a technology that bridges the gap between low-level events on the technical layers and business events on the application. CEP implementations like Esper, Tibco BusinessEvents and Progress Apama, to name a few, support this focus by providing a rich set of functionalities for the detection and derivation of complex events.

However, while the technology exists to derive high-level business relevant sense out of low-level runtime events, existing solutions favor bottom-up development strategies that focus on the processing of existing events. Complex events are derived and acted upon, but the development has a clear technical focus and the structure of the resulting Event Processing Network (EPN) is hard to align with the actual business goals.

For the best benefit of CEP there is a need for a goal-driven approach that facilitates different views on problem statement and CEP-based solutions (Figure 1).

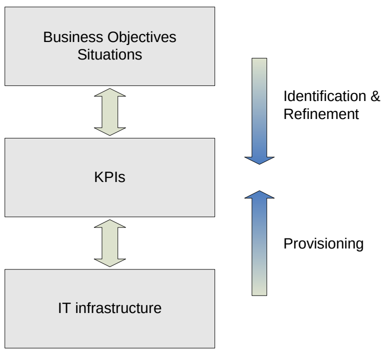

Figure 1: Viewpoints in CEP

Separating the analysis of situations and the definition of important KPIs helps analysts and engineers to concentrate on the right events and the most relevant situations. Such a focus also improves the maintainability of the EPN and allows for its lean implementation.

Additionally, a methodology is required to identify and refine the goals and KPIs necessary, together with their alignment to the data acquisition and processing in CEP applications.

This paper investigates a viewpoint-based structuring of an EPN and a methodology that supports a top-down development approach. Business-level events represent values for KPIs and situations relevant for the business are represented by the interpretations of these KPIs. As KPIs are more stable than business situations a separation of their modeling leads to better comprehensibility and increased maintainability. This separation has to be supported by the logical structure of an EPN.

The rest of the paper is structured as follows: The next section shortly outlines related work. Section 3 highlights the problems and challenges in the context of an industrial application example. The business-oriented structure of an EPN is explained in section 4 and the approach for a top-down and goal-driven implementation is presented in section 5. The last section presents ideas for the integration of CEP within an established Enterprise Architecture framework.

The terms related to event processing used in this paper have the meaning as defined by the Event Processing Technical Society (EPTS) in [5].

2. RELATED WORK

An introduction to CEP and its technical building blocks is provided by David Luckham in [10]. He describes the structure of an EPN as the interconnection of channels, Event Processing Agents (EPAs), producers and consumers. Sharon and Etzion present a meta-model for this technical structure in [15].

Regarding the design and engineering of such systems, Fowler and Hohpe et al. give a good overview [6][8] and Mühl et al. present technical foundations [11].

The business motivation for fast reactions to situations has been described by Hackathorn in [7]. In the context of active data warehouses he examines the relation of "business value to data freshness" and outlines the benefits achieved by shortening the decision latency for interpretation of events.

Prior to the existence of the term "CEP" Zachman and Sowa outline the relationship of temporal aspects to other perspectives of an enterprise architecture in [18] and [16]. They detail the importance of different perspectives and abstractions.

Within the research project "THESEUS" [17] the framework by Zachman has been successfully adapted to the domain of service engineering as described by Kett et al in [9]. Their research focuses on the identification of suitable notations for the individual cells, the vertical transformation between them as well as horizontal dependencies and a supporting methodology for a top-down development. Unfortunately, the "Time" perspective is currently not considered in the resulting Inter-enterprise Service Engineering (ISE) Framework, thus critical situations cannot be explicitly modeled and the engineering of event rules is not supported.

The support for complex events in business process notations has been recently examined in [1]. As a result of this research the new Business Event Modeling Notation (BEMN) is presented in [3].

3. CASE STUDY

We describe an exemplary case study from the manufacturing domain. It is obviously over-simplified in its technical details, but sufficient to highlight the challenges for CEP in industrial settings. Consider a car manufacturer. It runs several plants, which contain a number of production lines that mould and weld metal sheets, pre-assemble parts, varnish, etc. Some lines are replicated and produce the same type of artifacts, while others form a sequence of processing steps.

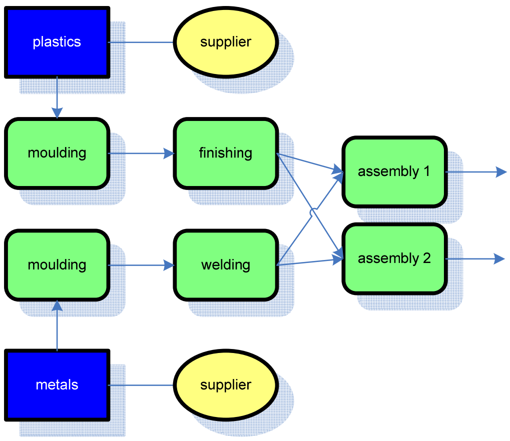

Figure 2: Simplified manufacturing process

The processes involve external suppliers, stretch over physically separated plants of the manufacturer, and consist of a wide variety of mechanical process engineering in each of the depicted steps.

In this paper, we consider the following scenario. The machines in the assembly line need regular maintenance, which leads to a shutdown of one line while the other continues to operate. This reduction of the overall utilization should be scheduled appropriately (e.g. during night shifts, etc.). However, the optimal point in time is not predictable in settings with non-constant material flows (e.g. quality rejections in discrete processes or varying processing times in continuous process technology). But what are the best situations to initiate a maintenance interval? What are the windows of opportunity to exploit?

In the above example this scenario may arise if one of the suppliers is not able to deliver the necessary amounts of material / in the desired quality. The processes running on the production lines within the plant will therefore not be able to continue in the planned manner. A number of compensation and contingency mechanisms may be applied in such a situation. For example:

- check stocks in other plants or in production lines for available material
- check for alternatives suppliers for the shortened material
- balance load between identical production lines
- concentrate available material on some production lines to stop others to start maintenance work

3.1 Automation Pyramid

In industrial settings the overall system architecture is often shown with the help of the automation pyramid. At the top level Enterprise Resource Planning (ERP) systems are responsible for high-level planning, stock management, and order handling. Management execution systems (MES) control the production processes in a plant, collect and archive KPIs and quality data. The next layer is about Supervision, Control, And Data Acquisition (SCADA). It controls and monitors specific production processes, which are run on the next lower level, and it collects the runtime data necessary for distilling KPIs on upper layers. On the next lower level Programmable Logic Controllers (PLCs) operate on actuators and sensor data to actually run the physical production process. At the bottom is the field level containing field buses and industrial communication networks.

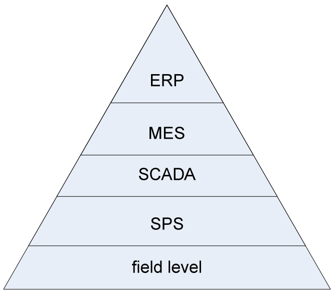

Figure 3: Automation pyramid

3.2 Challenges

The automation pyramid already shows that industrial application scenarios usually span several layers of abstraction, each comprising its own set of systems, languages, and engineering processes. However, typical CEP scenarios aim at enterprise applications, which are usually much more homogeneous, and a single (technological) view on to automation systems will hardly suffice. We see the following challenges:

- identify situations suitable for CEP
- define measurable objectives, i.e., target values for KPIs
- maintenance of event hierarchy understood by business analyst as well as process engineers
- methodology for a top-down engineering approach to derive KPIs, event types, and the necessary underlying event processing

KPIs enable us to supervise processes and are a means to support the desired business agility. As part of requirements engineering we need a thorough analysis of the business objectives to derive the KPIs and to link the objectives with the automation systems in later steps.

When these KPIs or their target values are changed by the management, often significant development efforts are induced. Maintaining the set of objectives, the KPIs and associated actions, the derivation of the necessary data is an important task in CEP application management. A clear distinction between the derivation of KPIs and their interpretation is a necessary separation of concerns, which facilitates implementation, maintenance, and communication between business and IT departments.

Even if involved manufacturing systems are developed largely independently and in iterations, a top-down engineering approach for the various increments on the layers of the automation pyramid is a valuable basis for a mature development process.

4. VIEWPOINTS TOWARDS CEP

To address the aforementioned challenges, we present a business-oriented structure of an EPN that clearly distinguishes the derivation of values for KPIs and the interpretation of these values in relation to business objectives. The situation to be detected is the opportunity to schedule a maintenance interval at the end of the production line (process step "Assembly" in Figure 2) if the utilization of the machinery will drop below 50%. This is indicated by reduced utilization at the beginning of the production line (process step "Molding").

We start with the derivation viewpoint for presentation purposes. In practice the development of the two viewpoints is done in parallel, proceeding from top to bottom of the automation pyramid in an iterative style, which is described in section 5.

4.1 Derivation

The derivation viewpoint is focused on the calculation of relevant KPIs. Here, it is the utilization rate. The Entity Relationship diagram shown in Figure 4 displays the relevant entities.

The term "Derived Event" is chosen as it is a more applicable to describe the abstraction process than the term "Complex Event" following the definitions provided by the EPTS in [5]. The derived event represents the value of a metric at a specific point in time; therefore the metric is equivalent with the event type of the derived event. The term "Metric" is used instead of "KPI" to indicate that several levels of abstraction are supported.

The values of a metric are usually stored in a sliding window. The dimension of this window is defined by the derivation rule of the upper level metric. A metric is a "virtual" event, generated by executing a rule that processes lower-level events [5]. These lower-level events can either also be derived events or simple events that are received from outside the CEP engine. The resulting abstraction hierarchy can be several levels deep and it ends if the lower-level events are represented by simple events that are sent to the CEP engine. The EPN from the derivation perspective therefore directly represents the metrics hierarchy of the business.

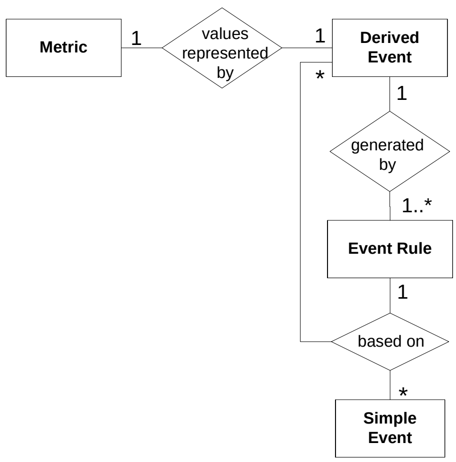

Figure 4: ER diagram for viewpoint "derivation"

The differences between this structure and existing approaches is that the structure shown in Figure 1 is business oriented as it is structured around a metric that is derived from business objectives. Existing researches are rather technical oriented. The structures of the resulting EPNs might be similar, but the intention of the implementation is more transparent following the top-down approach.

The technical orientation of an EPN is described by Luckham in [10] as well as Sharon and Etzion in [15]. They identify the following technical building blocks of an EPN: Event producers, event consumers, and event channels for interconnection. Event rules are contained in Event Processing Agents (EPAs) that are usually event consumers and producers at the same time. According to Luckham an EPA can be of three different types: (1) Filter, (2) map and (3) constraint. The type of an EPA represents the type of processing that is executed. The filter does not create new events, but just limits the events that are forwarded for further processing. Maps, however, represent the computation of derived events. The model in Figure 4 is simplified in this respect and does not differentiate between the two types and treats them both as "Event Rules". According to Luckham constraints differ from maps as they are focused on detection and generate violation notifications. This type of EPA is not required for the composition of an event abstraction hierarchy but is relevant for the interpretation viewpoint that is discussed in the next section.

4.2 Interpretation

The interpretation viewpoint focuses on the reactive part of the EPN: The generated notifications based on the detected state of the system that are used to trigger a reaction. This state is commonly defined as "situation". A situation that is to be detected using CEP has to be related to an objective of the business. Without such an identified objective, the discussion about KPIs and EPN would be arbitrary. An objective therefore represents the target value for a metric.

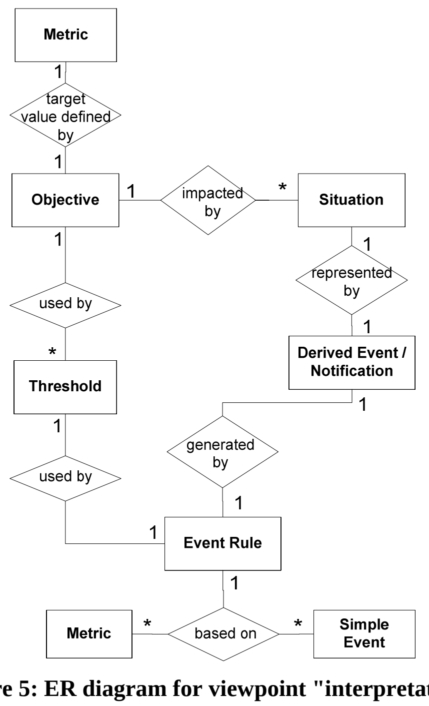

Figure 5: ER diagram for viewpoint "interpretation"

To avoid conflicts of interest there can only be a single objective per metric. But several situations can be defined based on the objective to support for example different severity levels of violations. A situation is represented by a derived event that has the semantic of a notification as it is generated to trigger actions in response to the situation. Similar to the derivation viewpoint the derived event is generated by an event rule that has now the same semantic as the EPA of type "constraint" [10]. The event rule also includes the entity "Threshold" that defines the critical value of the objective. Via several different threshold elements the same event rule template can be easily reused to model situations of different severity levels. The event rule is also based on other metrics and simple events.

The metric entities provide the well defined links between the two viewpoints.

4.3 Provisioning

As indicated in Figure 1 KPIs are based on information distilled out of the underlying, running system. From a data definition perspective we have to identify appropriate sources of information and the technical means to collect them at runtime. Lots of work exists that address this issue, ranging from adapters in enterprise application integration (EAI) to interceptors on network level and hooks on code level.

The more important perspective, however, is quality attributes of data and data delivery. The analysis of the situations, their priorities, and the nature of the KPIs is done to determine quality attributes like timeliness, auditability, idempotency, frequency, etc. of underlying events. This information is gathered during a top-down engineering (cf. Section 5) and drives the selection of architecture and technology, its sizing, and configuration.

For example, if detected situations and compensation actions affect the business relation to other partners, the exact definition of the situation and occurrences have to be recorded for non-repudiability reasons. As a consequence, the resulting system architecture contains a database for logging these kinds of data.

Timeliness and frequency may require dedicated communication lines and scalable message processing. Especially, in production processes we have different applications running on higher levels of the automation pyramid, which pose different requirements. Quality requirements for recording in product data management differ from those in utilization detection in manufacturing execution systems. This information, gathered in different, application-specific views, can be exploited to drive the engineering of the underlying systems here.

The approach to system engineering needs further details, of course, but we only sketch it here to highlight the benefits of different views in this paper.

4.4 Viewpoint Integration

The separation of the EPN into the two viewpoints addresses the requirements of maintainability and agility. It respects the different volatility of situations and their KPIs: Changes in situations to be detected can be handled without affecting the derivation of values for KPIs that are more stable.

For a tangible view on the EPN the two viewpoints need to be integrated. Since the metrics are key entities in both, they provide the well defined links between them.

Figure 6 shows the prototypical visualization component for the business oriented perspective towards an EPN. This component has been implemented within the research project "THESEUS" [17]. It unites the entities of the two ER diagrams and also displays the event instances for the metrics and notifications at runtime. The derivation viewpoint for the KPI is displayed in the upper part under the node "Derived by". It shows that that the processing is spanning several abstraction levels: The calculation of the utilization rate is based on the number of items that have been produced during the last hour at the molding process step. This metric is in turn derived from an event rule based on a one hour sliding window containing all events "ItemProcessed" that have been generated by sensors at the molding machinery.

The interpretation viewpoint is shown in the lower part under the node "Situations to be detected". In this case only a single situation has been defined for the metric. The interpretation event rule is based on the events for the KPI, and the threshold indicates that the notification is generated when the value of the utilization rate under-runs the threshold of 50% below the defined objective. The development of the viewpoints will be discussed in detail in section 5.2.

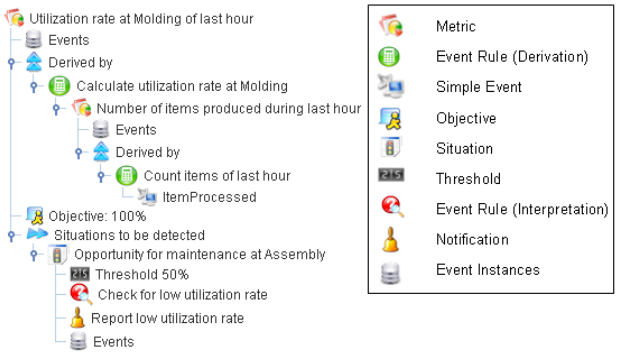

Figure 6: Visualization of business oriented EPN

5. METHODOLOGY

Before the EPN is modeled according to the structure outlined in the previous section, the targeted business goals have to be identified. Since CEP is not suited for every situation, this section first outlines how situations critical to the business are identified that are worth to be monitored with CEP. Then the top-down modeling of the EPN to detect the sample situation is explained.

5.1 Identifying Situations

The main characteristic of CEP is the fast detection of event patterns, enabling real-time reactions. In contrast to Business Activity Monitoring (BAM) it is not focused on the visualization and reporting of the current state of the business, but on the correlation of events over specific time intervals, i.e., the joining of event streams, the detection of complex event patterns, and the automatic and immediate initiation of further actions.

CEP is well suited for situations that need fast reactions. Hackathorn investigates in [7] the relationship of "data freshness to business value" using a value-time curve as displayed in Figure 7. It indicates how quickly the value of information about an event degrades over time and shows the value gained by reducing the latency until an action is taken.

Each situation needs to be analyzed with respect to this relationship: A large negative gradient of the "data freshness to business value" relation indicates that CEP can provide great benefits for the business by detecting the event and initiating the appropriate action immediately. A small negative gradient indicates that the event can be handled safely with a delay as the business does not benefit much from a real-time reaction.

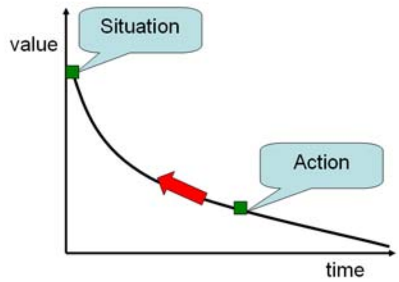

Figure 7: Relation of "data freshness to business value" cf. [7]

In industrial control settings the identification of the valuable situations is crucial because of the amount of data produced in the production processes. Without a goal-driven derivation of the required data, it would be unfeasible to query or even collect all the data and process it in real time.

In the presented use case the manufacturer's revenue depends directly on the produced items, therefore the revenue is directly correlated to the downtime of the machinery. The downtime can be reduced if windows of opportunities for maintenance intervals are detected. These situations need to be detected fast so that technicians can be informed and carry out the required preparations and actions.

5.2 Developing the EPN

After the situation "Opportunity for maintenance" has been identified using the value-time relationship it has to be formalized: First the main KPI and its objective that is affected by the situation have to be defined before the situation itself is represented by an event that is generated by an interpretation event rule.

In the sample use case the KPI is the average utilization over a one hour time window of the machinery at process step "Molding". For simplicity we assume that an objective of 100% utilization has been specified. The threshold for the situation is an average utilization of 50%, indicating that one of the machines of process step "Assembly" can be safely shut down. Again, this is simplified for illustration purposes. The interpretation event rule just checks if the current value of the KPI is lower or equal to the threshold value with respect to the objective. The generated notification is a simple event that can be further interpreted, for example by a rule engine, to decide which of the assembly machines should be shut down.

As a next step the derivation viewpoint is further detailed and the calculation (derivation event rule) of the utilization rate is specified. The values of the KPI are determined by relating the actual number of processed items of all molding machines for the last hour to the maximum possible number for a one hour period. The next lower-level metric is automatically indicated by this derivation rule: Number of produced items during the last hour.

This derivation of this metric's values is based on the simple events transmitted by sensors of the molding machines.

The number of abstraction levels required to derive the top-level KPI depends on the functionality provided by the used Event Processing Language (EPL) and the rest of the EPN structure: If no further situations need to be monitored for metrics in this derivation path, the derivation path can be more compact and the rules more complex. If further situations are defined and less complex rules are preferred, additional levels of abstraction should be used. Proceeding from top to bottom of the derivation path, the simple events become apparent that need to be received from the outside. If these events do not exist or do not contain the required data, sensors have to be included into the execution environments or explicit event generation has to be implemented into the applications.

This methodology ensures that only the events are processed by the CEP application that are relevant to detect situations.

6. INTEGRATION OF CEP WITHIN THE ENTERPRISE ARCHITECTURE

We see CEP as one discipline within an enterprise architecture, equally important as, for example, Business Process Management (BPM), Enterprise Data Management (EDM), and Business Rules Management (BRM), to name only some from the upper parts of the automation pyramid.

Situations detected using CEP cannot be treated in isolation: Events to be monitored are generated by process execution, metrics are based on data types, notifications trigger other business rules or processes, and there are several organizational units of the enterprise involved in the derivation process.

These interdependencies require a framework for the documentation and development of an enterprise architecture. One of the most prominent is the Zachman Framework [18], which has influenced many revised frameworks.

In the following, we highlight an approach to fit CEP into the Zachman Framework. Based on the research of existing specifications and notations for event rules, notations are identified that are seen as good candidates for the modeling of the EPN at the different perspectives of the framework.

6.1 Introduction to the Zachman Framework

The Zachman framework offers six different perspectives and abstractions for information systems, as shown in Figure 8. It divides the complex problem space in to manageable and well defined cells. This leads to a clear separation of concerns, and therefore to a better understanding of the system architecture and improves communication between stakeholders. The rows represent different stakeholders and their entries define the models used in these perspectives. The columns define different abstractions to the information system. While the framework itself does not imply any engineering methodology, we iterate from top to bottom for a goal-driven approach, refining the design with each layer. Dependencies between cells are only allowed amongst abstractions within the same row and to the adjacent abstractions in the same column. By prohibiting diagonal dependencies the impact of changes within a cell becomes manageable. Without this restriction the number of relationships between cells is too complex to be controllable.

	Data *What*	Function *How*	Network *Where*	People *Who*	Time *When*	Motivation *Why*
Objective/ Scope (contextual) → *Role: Planner*	List of Things important in the Business	List of Core Business Processes	List of Business Locations	List of important Organizations	List of Events	List of Business Goals/ Strategies
Enterprise Model (*Conceptual*) → *Role: Owner*	Conceptual Data/ Object Model	Business Process Model	Business Logistics System	Work Flow Model	Master Schedule	Business Plan
System Model (*Logical*) → *Role: Designer*	Logical Data Model	System Architecture Model	Distributed Systems Architecture	Human Interface Architecture	Processing Structure	Business Rule Model
Technology Model (*Physical*) → *Role: Builder*	Physical Data/ Class Model	Technology Design Model	Technology Architecture	Presentation Architecture	Control Structure	Rule Design
Detailed Representations (*Out of Context*) → *Role: Programmer*	Data Definitions	Program	Network Architecture	Security Architecture	Timing Definition	Rule Specification
Functioning Enterprise → *Role: User*	Usable Data	Working Function	Usable Network	Functioning Organization	Implemented Schedule	Working Strategy

Figure 8: Zachman Framework cf. [16]

Since 1992 the nomenclature of the Zachman Framework has been subject to changes, for example the layer "Detailed Representations" has been renamed to "Component Layer" and the perspective "Function" is now called "Process".

By inspecting the labels of the cells we see that the "Time" abstraction is well suited to contain the modelings defining the EPN: It deals with the temporal relations of events and therefore matches the focus of CEP.

6.2 Models for Layers

We propose different notations and specifications that can be used to model the EPN in the column "Time" according to the detail level required for the perspectives of the Zachman Framework.

6.2.1 Contextual Layer

At the contextual layer the situations are identified that have the greatest impact on the business goals and objectives. For a goal-driven approach to CEP it is therefore mandatory that the business motivation is captured. The OMG has developed the Business Motivation Model (BMM) specification [12] for this purpose. An introduction to this model including a case study has been provided by The Business Rules Group [2]. The BMM is not focused on events and can be used to structure the information across all abstractions of the top-level perspective of the Zachman Framework.

Figure 9 depicts the main elements of the BMM. Desired results, "Ends", are specified by long-term goals and quantified by short-term objectives. Strategies and tactics, "Means", define how the goals and objectives are planned to be achieved. Both ends and means are related to influencers via assessments. In the example use case the objective "Revenue per month of $100000" has the internal influencer "machinery". The assessment analyses the company's Strengths, Weaknesses, Opportunities and Threats (SWOT) and includes a potential impact analysis with respect to the defined objectives and goals. The varying flow of material is a weakness and the dynamic scheduling of maintenance is an opportunity.

The impact analysis should include the evaluation of the value-time relation as described previously. Additionally other non-functional requirements need to specified, e.g. the auditability and traceability of events that later affect the technical architecture of the CEP application.

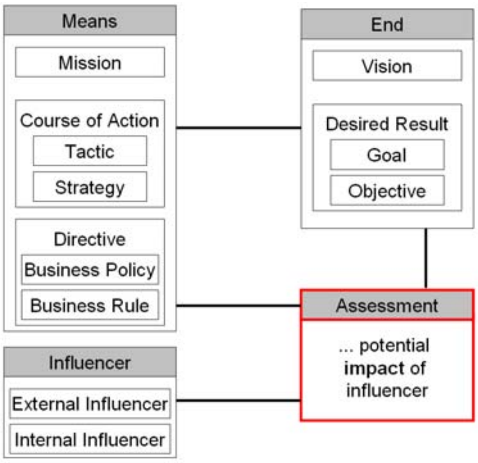

Figure 9: BMM overview cf. [12]

6.2.2 Conceptual Layer

At the conceptual layer the situations are refined and modeled in a formal notation. Barros et al. create in [1] a requirements catalog to model complex events in process models and investigate the process modeling specifications BPMN 1.0 and BPEL 1.1. Based on this gap analysis they present in [3] the Business Event Modeling Notation (BEMN). The corresponding meta-model is shown in Figure 10. BEMN allows to graphically model complex events that can be linked to business processes designed with BPMN. Decker and Barros show that BEMN is suitable to model the basic event patterns despite this rather small set of constructs.

Similar to BPMN these patterns are not meant to be directly executable. BEMN currently does not provide constructs to differentiate between the viewpoints "derivation" and "interpretation".

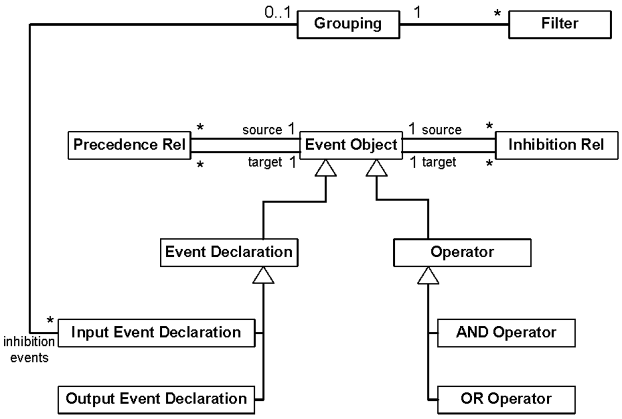

Figure 10: BEMN meta-model cf. [3]

6.2.3 Logical Layer

The purpose of the logical layer is to provide models that can be the basis for transformations to executable code. This approach is already well established at the data abstraction to generate implementation classes for different target platforms out of UML class diagrams.

The model of the conceptual layer should be refined with exact temporal relations of the events. Timing diagrams have been introduced to the UML 2.0 specification [14] that allow the modeling of temporal relations but lack the capability to use logical operators or groupings. Therefore this notation would have to be extended to include at least the elements of the BEMN in order to be suitable to model event rules.

To the knowledge of the authors currently no suitable notation exists that fulfills the requirements of modeling event patterns on the logical layer. The Object Management Group (OMG) is currently working on a new specification, the Event Metamodel and Profile (EMP) [13], which aims to provide the required functionality. It is currently an RFP status.

6.2.4 Physical Layer

This layer contains the exact definitions of events and their structure in the resulting EPN. Currently several textual Event Processing Languages (EPLs) exist like RAPIDE [10] or EsperTech's Esper EPL [4] that can be used to implement an EPN following the viewpoint clustering presented previously. These EPN implementations are bound to a specific runtime since the EPLs are designed for a single target platform.

6.2.5 Component Layer

At the component layer the actual deployment of an EPN in the system architecture is specified. The system architect needs to investigate and estimate the resources needed by the individual patterns to decide how to distribute them across the available CEP engines.

Besides extra capacity for the CEP engines, the system architecture must be adapted to cope with extra network load. Capacity and load models have to be adapted accordingly, too.

KPIs and the necessary underlying data have been identified in the derivation view, which is taken up on the logical layer of this framework. On the component layer, we have to identify the components containing the necessary data, appropriate probes and data extractors have to be provided, and an additional data handling and messaging infrastructure might be necessary. While UML diagrams are often used on MES layer of the automation pyramid, their ability to express data extraction views on base data is rather limited. In custom data models (or UML profiles), which are often used on lower levels of the pyramid, it is easier to express extraction views.

As long as we are dealing with enterprise repositories and databases, many existing solutions provide the necessary connectors and communication facilities. In the context of industrial automation, we expect more custom solutions because of the heterogeneity of systems.

6.3 Outlook

The presented models are only a first starting point to support different perspectives on an EPN. While standards exist for models of other columns like "Data" and "Process" (cf. [9]) only the BMM could be identified as an existing official specification that is partially suitable for the "Time" abstraction. Additionally, none of the investigated notations support our proposed viewpoint structuring. Significant developments and standardization efforts are yet to be done to achieve an integrated development of the EPN that follows the perspectives of the Zachman Framework and is integrated with models of the other perspectives.

7. CONCLUSION

This paper has outlined the motivation and benefits of using CEP in the automation industry. Fast reactions to critical situations make a company and its processes more adaptive and responsive to changes. In the context of industrial automation, we distinguish different viewpoints on CEP applications and present an approach for top-down development within the overall system architecture.

In order to cope with the wealth of systems and data on the various layers of the automation pyramid, we propose the structuring of an EPN, which is built around the important metrics of a business. A top-down definition of objectives and of the necessary KPIs is separated in the interpretation and derivation viewpoints. They focus i) on metrics and the event rules that represent the situations to be detected and ii) on how the values of the high-level metrics are computed from low-level simple events generated by the IT systems, respectively.

This approach to CEP engineering has been embedded into an enterprise architecture framework. The "Time" perspective of the Zachman Framework is well suited to contain models of the EPN at different perspectives.

The resulting structure makes EPNs better to understand, because it explicitly documents the supported business objectives. It improves maintenance, because stable entities, like KPIs and their computation, are separated from volatile entities, situations and thresholds. The presented work is an initial summary of applying CEP within the automation domain. Further measurements must corroborate the good experiences in other projects. Future research is necessary to extend the models for CEP, investigate possible model transformations between perspectives and the dependencies to other abstractions.

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